Microring modulators.

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Microring modulators

working principle

-Resonance occurs when light circulating inside a ring-shaped waveguide constructively interferes with itself.

-This happens only under a specific condition :The optical path length of the ring must be an exact multiple of the light’s wavelength[2].

-When the waves in the loop build up a round trip phase shift that equals an integer times 2π, **Φ = 2πm** , the waves interfere constructively and the cavity is in resonance.

Φ = βL is the single-pass phase shift, with L the round trip length and β the propagation constant of the circulating mode.

$$\lambda_{\mathrm{res}} = \frac{n_{\mathrm{eff}} L}{m}$$

Microring modulators

working principle

-modulating the optical output at a fixed laser wavelength. In practice, index tuning is achieved by:

-Carrier-induced effect (Si) “plasma-dispersion”

-Electro-optic (Pockels) effect

-Thermo-optic effect

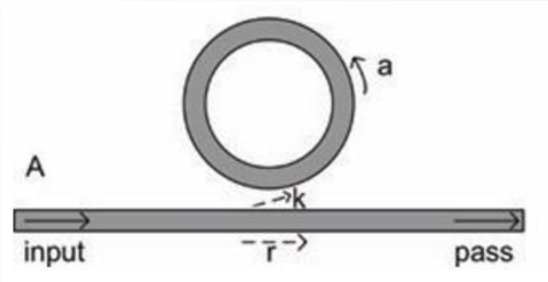
Microring modulators

Transfer function

-An all-pass ring resonator has one straight bus waveguide coupled to the ring.

-Assuming continuous wave (CW) operation and matching fields. Under the assumption that reflections back into the bus waveguide are negligible, we can write the ratio of the transmitted and incident field in the bus waveguide as:

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = e^{i(\pi+\phi)} \frac{a - re^{-i\phi}}{1 - rae^{i\phi}}$$



φ is the single-pass phase shift, a is the single-pass amplitude transmission, including both propagation loss in the ring and loss in the couplers. It relates to the power attenuation coefficient α [1/cm] as: $a^2 = \exp(-\alpha L)$.

Microring modulators

Transfer function

- By squaring this Eq., we obtain the intensity (power) transmission Tn:

$$T_n = \frac{I_{pass}}{I_{input}} = \frac{a^2 - 2ra\cos\phi + r^2}{1 - 2ar\cos\phi + (ra)^2}$$

- I is The optical intensity, r is the self-coupling coefficient. Similarly, we can define k as the cross-coupling coefficients, and so r^2 and k^2 are the power splitting ratios of the coupler, and they are assumed

to satisfy $r^2 + k^2 = 1$, which means there are no losses in

the coupling section.

Microring modulators

Footprint

-The microring modulator is known to exhibit small footprint.

-tens of μm circumference.

-A smaller footprint → more devices per chip → higher integration density.

Speed

-high-speed operation.

-The operating speed of the modulator is limited by the charging time of the p-n junction, and the photon lifetime in the resonating structure

Microring modulators

Extinction ratio

-A sharp resonance (high Q factor) is required to achieve high dynamic extinction ratios with a relatively small resonance shift. However, a high-Q resonator has a longer photon lifetime which limits the achievable modulation bandwidth.

Power consumption

-the relatively small area limits the necessary power to modulate. Therefore, ring modulators currently hold the best track record in terms of modulation energy per bit[2].

-A small resonance shift only requires low driving voltages and, therefore, achieves low power consumption by the modulator.

Microring modulators

-cascaded microring-based silicon modulator, where The cascaded double ring structure shows advantages of a large optical bandwidth and a steeper sideband over a single ring with large extinction ratio[1].

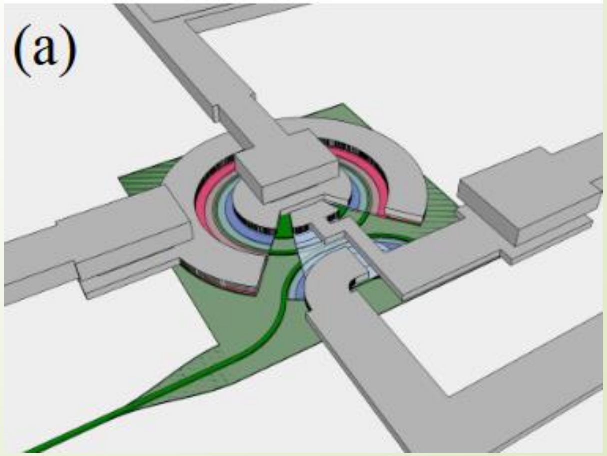
-High-speed modulation up to 40 Gbit/s with extinction ratio of 3.9 dB was also experimentally achieved at drive voltage of 5 V. A high extinction ratio of 8.6 dB was reached at 20 Gbit/s.

Microring modulators

-An O-band modulator based on silicon microring devices. [3]

-The modulator has 25 dB extinction ratio (ER) and ~2dB on-chip loss, as shown in Fig. 3(a). The on-chip propagation loss is ~1 dB, and the insertion loss of the modulator is ~1 dB, and operates at 40 Gb/s near 1310 nm where SMF exhibits minimal dispersion in the O-band.

(a) Layout of the modulator. Light green stands for the 90nm thick Si, dark green for the waveguide, and gray for the back-end metallization. Red stands for p doping, and blue for n doping. Note: deeper the color, heavier the dose.



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- [2] Bogaerts, W., De Heyn, P., Van Vaerenbergh, T., De Vos, K., Kumar Selvaraja, S., Claes, T., Dumon, P., Bienstman, P., Van Thourhout, D. and Baets, R. (2012), Silicon microring resonators. Laser & Photon. Rev., 6: 47-73. https://doi.org/10.1002/lpor.201100017
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